

Reproducible Wildfire Exposure Screening for Residential Location Selection in Mainland Portugal

Historical screening plus a validated annual comparative-exposure model—not a property-level safety forecast.

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Historical evidence compares broad areas—not properties

How can historical wildfire evidence help compare candidate residential areas in mainland Portugal?

The decision is national comparative shortlisting: apply one transparent method before investing time in detailed municipal, site, and property checks.

MAINLAND PORTUGAL

89,112

validated grid cells

1 km

sole analytical cell resolution

Scope boundary: historical comparative screening only—no property-level safety guarantee or purchase decision.

National coverage makes the method consistent

and supports honest temporal validation

NATIONAL FIRST

One method across the full range of mainland wildfire conditions

89,112 cells and 15 labelled predictor years supported strict temporal evaluation before a fixed nine-feature model was refit through 2025 outcomes.

Consistent

The same definitions and processing rules apply across mainland Portugal.

Representative

The screening spans the country's broad range of historical wildfire conditions.

Testable

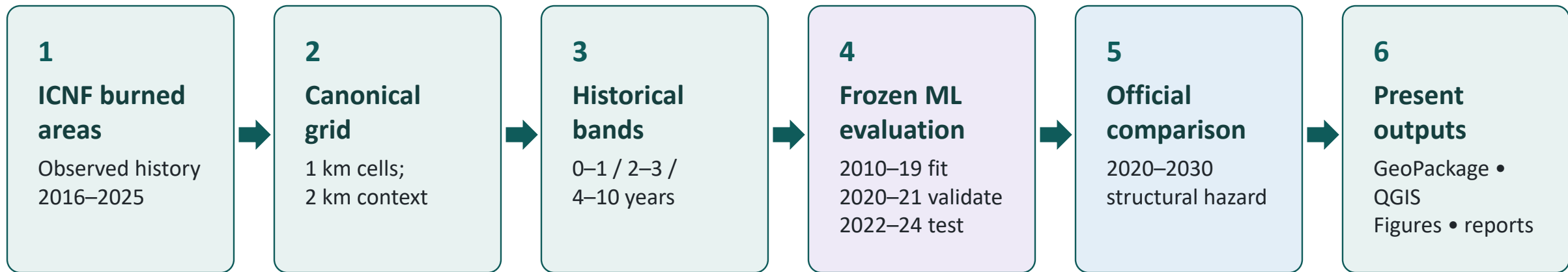
Separated training, validation, and final-test years support an honest temporal evaluation.

Reusable

The result is one inspectable national GeoPackage and QGIS presentation layer.

National screening does not replace municipal or property-level assessment.

One workflow keeps three evidence roles distinct



Nine-feature model context

CLC land cover • Copernicus DEM terrain • ERA5-Land JJAS climate • strictly prior fire history

The official ICNF structural-hazard layer is neither the regression target nor an accuracy benchmark.

Zero-heavy outcomes make low error deceptively easy

Exact zeros dominate the target

A regression that stays near zero can achieve a low average error while failing to rank many fire-affected cells near the top.

0

exact zero is a valid outcome

Three complementary questions

MAE

How large is the average absolute error?

RMSE

How strongly are larger errors penalised?

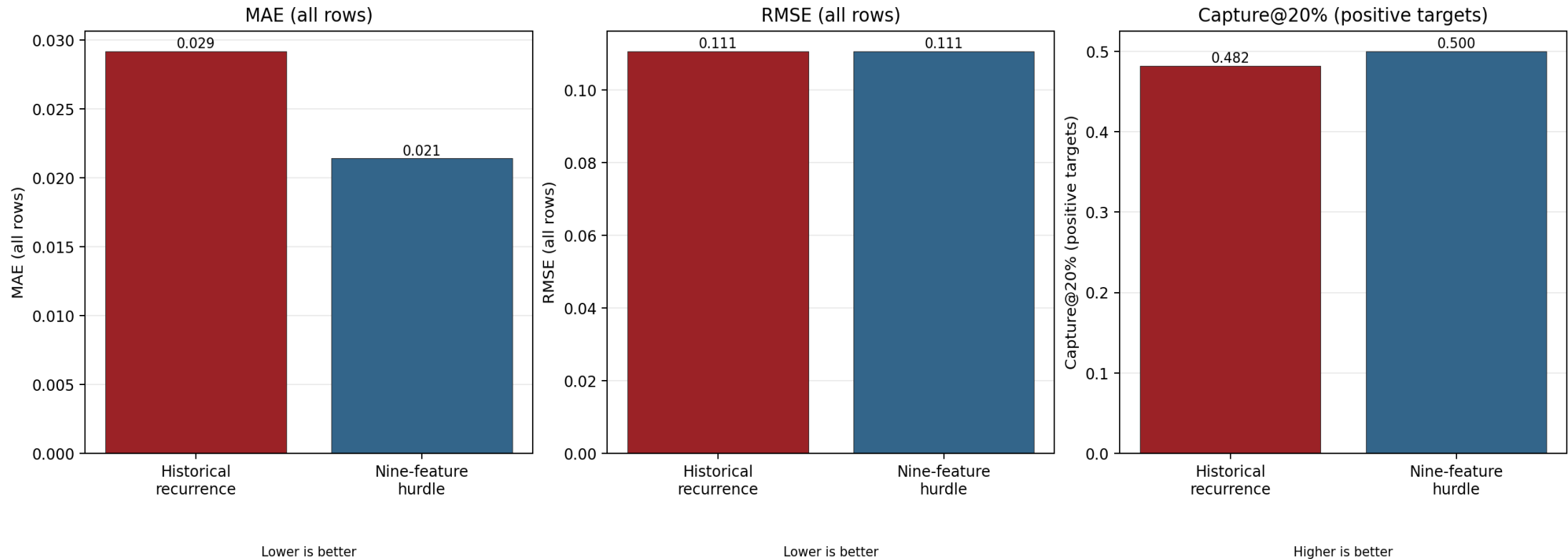
Capture@20%

How many positive-target cells appear in the top-ranked fifth?

Comparative estimates require error, calibration, and ranking evidence.

Nine-feature hurdle: lower MAE, better ranking

Frozen final-temporal comparison (T=2022-2024)



The hurdle lowers all-row MAE and improves capture@20%, while RMSE is similar and high-burn years are underpredicted. It is retained for cautious annual comparative estimates; historical recurrence remains descriptive context.

On T=2022–2024, the hurdle reduced MAE and improved Capture@20% versus recurrence, while RMSE remained similar and the high-burn T=2024 outcome was underpredicted.

From buyer request to comparative location shortlist

1 Buyer request

Compare broad candidate areas before detailed property checks.



2 Reproducible lookup

Read observed historical exposure and the complementary official class consistently.



3 Appropriate next step

Prioritise local due diligence according to the evidence—not an automatic accept or reject.

Fictional example — not a property recommendation

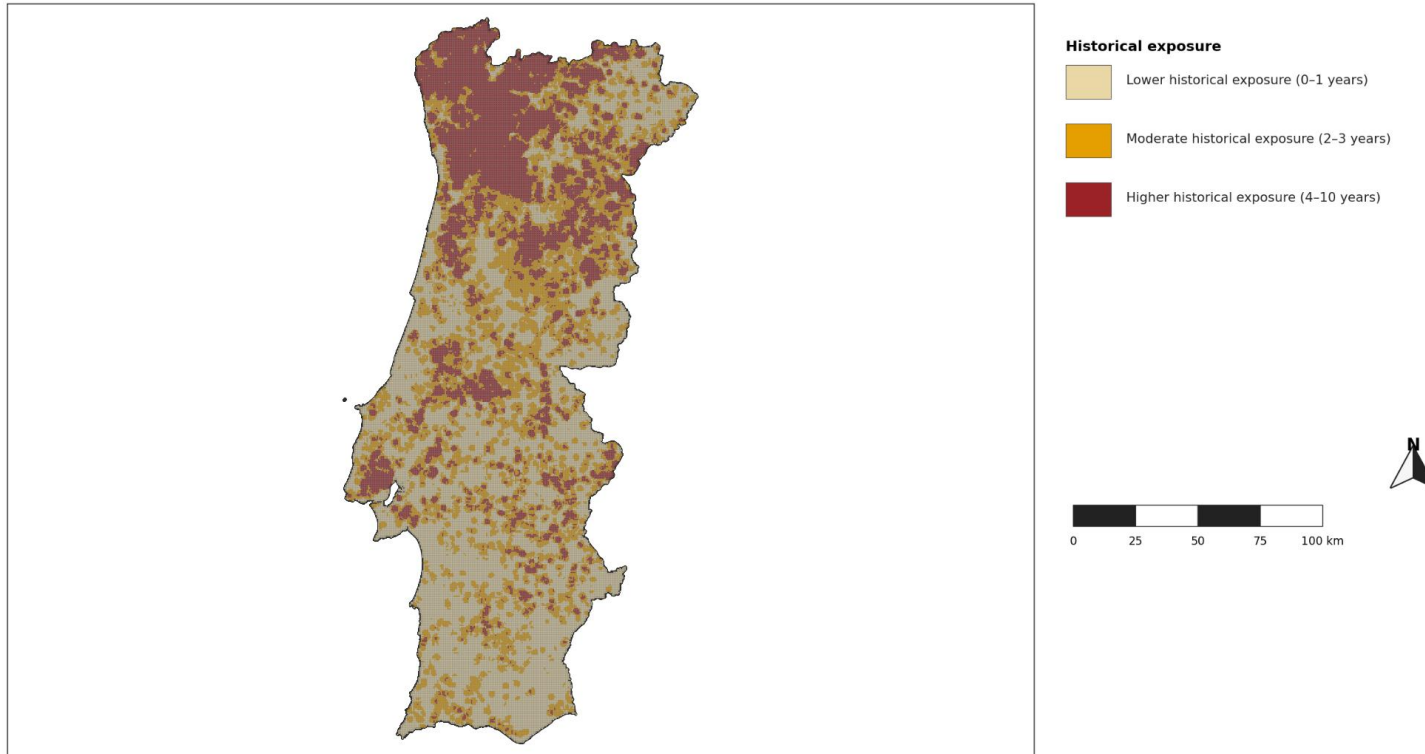
Candidate area	Historical exposure	Official structural hazard	Appropriate next step
Area A	Lower	Moderate	Prioritise detailed local checks
Area B	Moderate	High	Investigate prevention and access conditions
Area C	Higher	Very High	Require deeper specialist and local due diligence

The final classes summarise observed historical evidence consistently across the country. They support comparative shortlisting, not a property-level prediction or purchase decision.

Observed recurrence defines the final screening bands

Historical Wildfire Exposure Screening — Mainland Portugal

Historical exposure bands for 1 km mainland cells

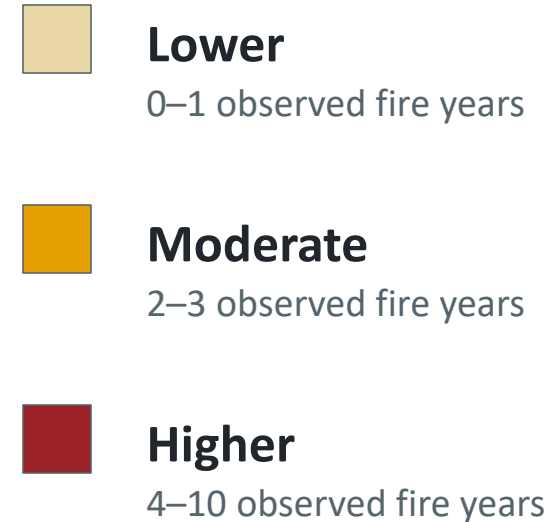


EPSG:3763 | Sources: CAOP 2025 boundary; ICNF annual burned areas; ICNF structural hazard map 2020–2030.

1 km cells; recurrence measured in 2 km context; 2016–2025 evidence

Historical comparative exposure only; not a next-year forecast, property-level safety guarantee, or purchase recommendation.

OBSERVED 2016–2025 EVIDENCE



1 km mainland grid cells with fire recurrence measured in a 2 km context.

Lower historical exposure does not mean safe.

Most cells fall in lower or moderate historical exposure

74.70%

of mainland cells are in the lower or moderate historical-exposure bands



Lower

36,645 cells

41.12%



Moderate

29,919 cells

33.57%



Higher

22,548 cells

25.30%

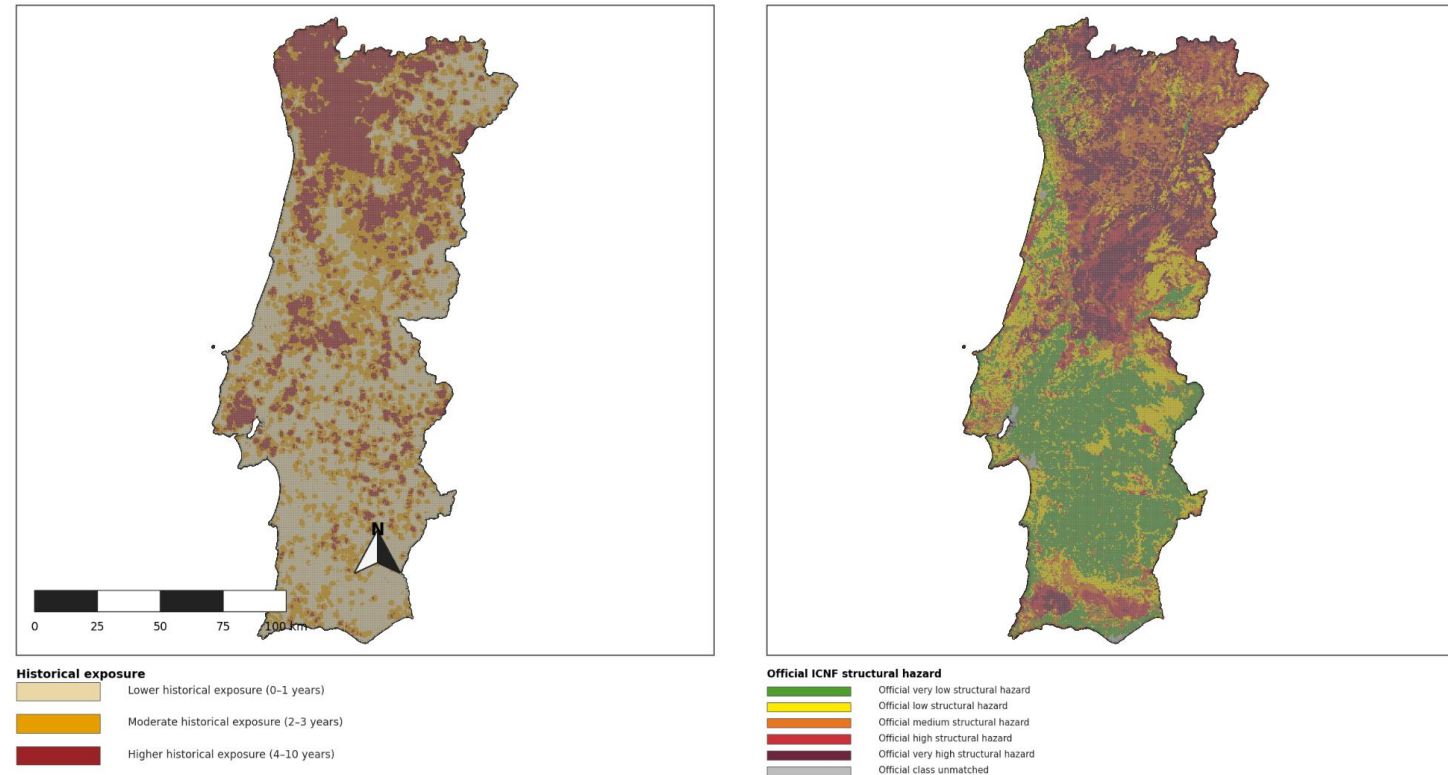
Complete national grid: 89,112 cells. Distribution is descriptive—not predictive performance or map accuracy.

The two maps answer different questions

Observed recurrence and official structural propensity are complementary

Historical Exposure and Official ICNF Structural Hazard — Comparison

Left: historical recurrence bands. Right: official ICNF structural-hazard class, summarized to 1 km cells.

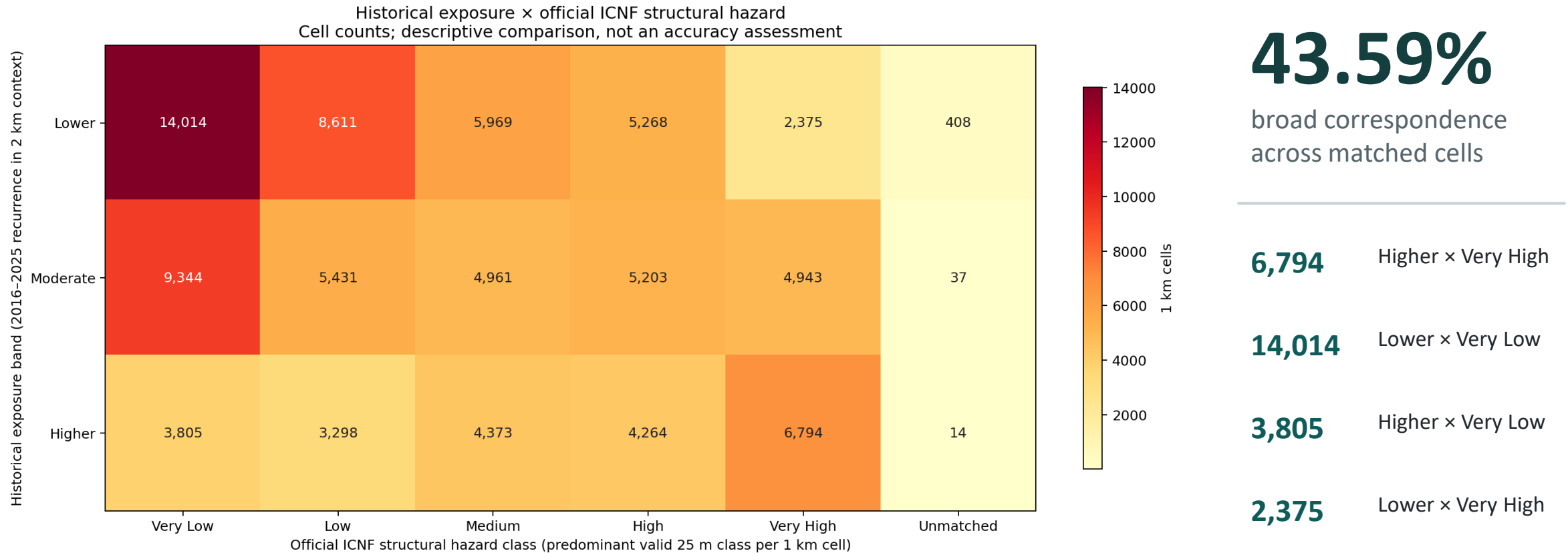


EPSG:3763 | Sources: CAOP 2025 boundary; ICNF annual burned areas; ICNF structural hazard map 2020-2030.

1 km cells; recurrence measured in 2 km context; 2016-2025 evidence

Historical comparative exposure only; not a next-year forecast, property-level safety guarantee, or purchase recommendation.

Correspondence is informative—not an accuracy score



Descriptive correspondence only—not ICNF-map accuracy or model accuracy.

Two complementary outputs, explicit limits

EVIDENCE-TO-USE CONTRACT



Use: broad location comparison and shortlisting; compare separately with official ICNF structural hazard.

Not a property-level safety guarantee, wildfire-risk probability, insurance estimate, or purchase recommendation.

CONCLUSION

- 1 Historical recurrence remains the transparent reference and a separate observed-history screening layer.
- 2 The frozen nine-feature hurdle improved final-test MAE and ranking, with similar RMSE and cautious high-burn calibration.
- 3 The result combines an inspectable historical GIS layer with a reproducible annual comparative-estimate workflow.

ACTUAL DELIVERABLES

Python project + numbered notebooks • validated screening GeoPackage • portable QGIS project • figures, tables and validation reports
• final PowerPoint presentation

Interactive layer: [qgis/wildfire_exposure_screening_portugal.qgz](#)

Likely assessor questions

Why all mainland Portugal?

The buyer's first comparison is national; local property checks follow after shortlisting.

What was the hypothesis and result?

The frozen nine-feature hurdle improved MAE and ranking versus recurrence, but not RMSE, on the final temporal test.

Were the features sufficient?

Nine governed features supported a useful comparative model; calibration remains weak for unusually high-burn years.

Was the project successful?

Yes: it produced validated historical screening and a tested annual comparative-estimate workflow with explicit limits.

Why is the classification still useful?

It directly summarizes observed 2016–2025 recurrence for consistent broad-location comparison.

Why retain recurrence if it is not prediction?

It remains an interpretable observed-history reference, separate from the model's annual estimate.

Central conclusion: retain both products—historical screening for context and the fixed hurdle for cautious annual comparison.